
Narrowing Without Converging: How Interactive LLM Agents Search Under Feedback

Anonymous Author(s)

Affiliation

Address

email

Abstract

1 An interactive agent searching under feedback should contract toward the answer.
2 These agents contract and arrive nowhere. We instrument Turtle Soup, a lateral-
3 thinking game of yes/no questions, with two process-level probes — a forced
4 and scored checkpoint story every round and an association trajectory embedded
5 from each round’s question — and run three Qwen tiers on 22 human-played
6 puzzles. Interaction helps for ten of thirty rounds and then stops, at every tier,
7 and neither the round budget, the Oracle, nor puzzle difficulty accounts for it.
8 The per-round step contracts four to five fold over the same rounds, so the search
9 narrows while accuracy does not move. Three explanations are tested in turn. The
10 answer barely redirects it: controlling for round position, the smallest tier’s next
11 move is statistically independent of whether it was just confirmed or refuted (Δ
12 stride +0.001, 95% CI $[-0.009, +0.011]$), while the larger two move *further* after
13 a confirmation (+0.028, +0.009), and a test that needs no premise about step size
14 shows all three moving toward regions the Oracle has already refuted, the smallest
15 tier most strongly. And the agent does not stop when it finds the answer: the largest
16 tier reaches a solution-grade story in six games and volunteers one, ending with a
17 story that sits no further from the one that scored 1.00 than any two rounds of the
18 same game sit from each other, so the moment of loss is invisible to the geometry
19 that describes it. One endpoint number scores all of this identically. We release
20 the harness, instruments, puzzle set, and per-game logs.

21 1 Introduction

22 Here is a Turtle Soup puzzle. *A man’s body lies in the desert, clutching half a matchstick; luggage*
23 *and clothing are scattered around him.* That is all the solver sees. The hidden story: a hot-air
24 balloon was losing altitude; the passengers threw out their luggage, still could not stay up, and drew
25 lots with matchsticks; the man who drew the short one was thrown out. The solver recovers this
26 story by asking yes/no questions, one per round.

27 Outcome metrics can say whether an agent got there; on our set that story is flat, and the round
28 budget is not the constraint (§3). What they cannot say is how the agent searched: one zero-scorer
29 never leaves its first hypothesis, another finds the answer and lets it go. What the process shows
30 is a search that contracts without converging. Turtle Soup suits the job: the behavior *is* text, the
31 Oracle’s reply is a clean one-trit signal whose effect on the next move can be isolated, and puzzles
32 are deliberately under-determined. Prior turtle-soup benchmarks score outcomes, not the search’s
33 geometry (Appendix A) [1, 2, 3].

34 An interactive agent is supposed to *use* what it is told, so we ask first whether it does: does the
35 answer just received change what the agent does next? Embedding each round’s question makes
36 this measurable, and the answer is this paper’s spine: barely, and where it does, backwards. Three

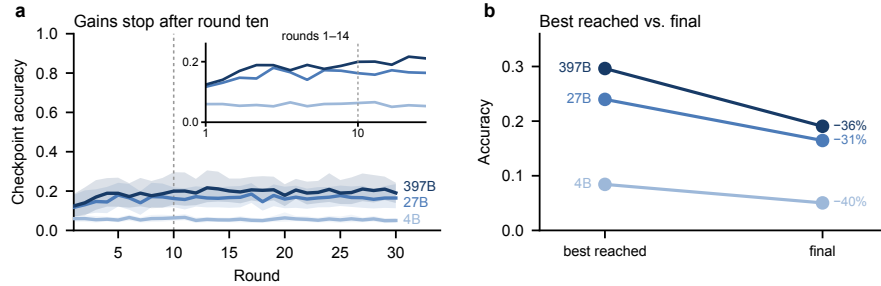


Figure 1: (a) Accuracy of the answer each tier would give at each round (bands, 95% cluster-bootstrap CIs over 22 puzzles); dashed rule at round 10; inset, rounds 1–14. (b) The best answer each game reached, against the answer it finally stands on.

37 questions follow. **H1**, the search narrows in on the answer as “no” answers eliminate hypotheses, so
 38 a shrinking stride means closing in. **H2**, the answer redirects the next question: a refutation should
 39 push the search away, a confirmation should refine it. **H3**, when the agent reaches the answer it will
 40 stop — the game ends at its peak. §3 takes them in turn.

41 **Contributions.** (1) A characterization of interactive search that **narrows without converging**: the
 42 per-round step contracts four to five fold at every tier while accuracy stays flat, and the feedback
 43 that should redirect it does not — stride conditioned on the answer received, with round position
 44 controlled, is absent at the smallest tier and inverted at the larger two, and told “no” every tier
 45 moves toward what it has already ruled out. (2) Two failure patterns **derived from it rather than**
 46 **catalogued**, circling and abandoning, with per-game criteria and counts. (3) An honest boundary:
 47 the geometry describes these failures but cannot score them — losing the solution is no larger, in
 48 embedding space, than ordinary within-game motion. (4) Released harness, instruments, a 22-puzzle
 49 human-played set, and per-game logs.

50 2 Framework and experiments

51 **Task and harness.** A **Questioner** sees only the *surface* (the puzzle’s visible scenario) and asks
 52 one yes/no question per round; the interactive player–judge protocol follows SPLAT [2]. An **Oracle**
 53 holds the hidden story and answers yes, no, or irrelevant. The Questioner may commit a final story
 54 at any round, or is forced to at the budget. The open-source harness logs every round, with per-game
 55 seeds.

56 **Checkpoint and commitment instrumentation.** Our first instrument does not wait for the end:
 57 every round, after the Oracle answers, the Questioner must write its best complete story, and *that*
 58 checkpoint is scored; each point on an accuracy curve is therefore a full answer written with the
 59 evidence available at that round. Alongside it we record *commitment*: whether and when the agent
 60 ever volunteers a final story of its own accord. Together these turn an endpoint benchmark into a
 61 scored process.

62 **Puzzles.** A puzzle nobody has solved carries no evidence that its clues suffice or that the intended
 63 leap is reachable, so our set takes only puzzles with a public record of human play: 22 items from
 64 a Chinese Turtle Soup community site, each keeping its source URL (surfaces 9–108 characters,
 65 solutions 13–200, 97 annotated key clues). Provenance is enforced in code, with a verified family, a
 66 quarantined generated family, and tests that fail if the two mix.

67 Each puzzle also carries an **under-determination** score — how far the surface alone leaves a solver
 68 from the solution, the closest of twelve cold guesses written with no Oracle feedback — used in §3
 69 as a puzzle-level covariate (distribution and properties in Appendix B).

70 **Outcome scoring.** Judged outcome is **clue recall (70)** plus **causal-logic identity (30)** (detail in
 71 Appendix C): string matching alone cannot tell a wrong answer from a right one worded differently
 72 — we watched a frontier model reconstruct a hidden causal chain completely and score 0.00 — and
 73 the clue half blocks “far = creative” gaming, because distance without validity scores zero.

Association-trajectory instrumentation. The second instrument reads the questions themselves, adapting forward flow [7] and the Divergent Association Task [8] from human psychometrics. Each round, we extract the question’s keywords (jieba), embed them (bge-small-zh-v1.5), and average them to a unit round vector q_t ; distances are cosine, and anchors embed the puzzle’s surface, solution, and clue terms (specification in Appendix C). Two signals follow: the **stride** $s_t = d(q_t, q_{t-1})$ and the **anchor distance** $h_t = d(q_t, \mathcal{A})$, with H1’s signature $\bar{s} \rightarrow 0$ and H2’s a rising h_t ; the lexical `question_novelty` metric is the ablation baseline that H1 predicts will miss synonym drift.

Designs. E1 (round curve) plays to 30 rounds under the checkpoint instrument. E2 (round budget) imposes hard caps $\{5, 10, 15, 20, 25, 30\}$ with a forced final answer only, so it doubles as a no-checkpoint control. E3 (association trajectory) computes $\{s_t\}$ and $\{h_t\}$ for every E1 game and answers H1–H3; the keyword extractor is fixed across models, so none is scored through its own vocabulary.

3 Results

All numbers come from one grid on the verified set: 22 puzzles $\times$ {Qwen3.5-4B, Qwen3.6-27B, Qwen3.5-397B-A17B; hereafter 4B, 27B, 397B} $\times$ 3 seeds (E1: 198 thirty-round games; E2: 1,188 across the six caps); accuracy is the composite score rescaled to $[0, 1]$. **Conditions of validity, in one sentence:** the Oracle and logic judge are DeepSeek-V3.1, cross-family with the Questioners and audited at 95% on held-out probes, and one environment artifact (token exhaustion at the smallest tier) is removed below before any behavioural comparison, with protocol and settings in Appendix C. The three models differ in generation and architecture (397B is a 17B-active mixture-of-experts), so we speak of model tiers, not scale; harness failures were zero.

What the plateau is not. Interaction helps for ten rounds and then stops (Figure 1a): the round 10 to 30 change is $-0.009 / +0.003 / -0.013$, CIs spanning zero, and each tier ends below the best answer it reached (Figure 1b). It is not the round budget: end accuracy is insensitive to the cap at the larger tiers, and the smallest model’s apparent decline is token exhaustion, rising 0/66 to 63/66 across caps with no decline among non-exhausted games. It is not a degenerate Oracle, audited at 95% with replies running 68–74% “no”. And it is not the puzzles: under-determination predicts what a model writes before any evidence arrives ($\rho = -0.34 / -0.54 / -0.47$) and nothing after it — not the round 10 to 30 change, any trajectory indicator, either half of the score, or the gap between tiers (all n.s. at $n = 22$, bounds rather than absences; Appendix C). The evidence is there, the budget is there, the puzzles are within reach, and the agent stops making progress anyway.

H1: the search narrows in on the answer. *It narrows; it does not converge.* Accumulated “no” answers should contract the search around what survives. The contraction is real: mean stride falls monotonically at every tier, four to five fold, from 0.145 / 0.232 / 0.182 at round 1 to 0.025 / 0.053 / 0.045 at round 29. The convergence is not. Accuracy over those same rounds is flat (§3), and within a tier stride does not predict how a game ends ($|\rho| \leq 0.16$). The narrowest searcher of all — 4B, at a late-game stride of 0.030 [0.021, 0.039] against 27B’s 0.082 [0.065, 0.098] — is the one that never reaches a solution-grade answer. Narrowing is what circling looks like from outside: orbits of one hypothesis basin, tightening around nothing. Not a malformed-question artifact: excluding 4B’s 17% truncated or non-interrogative questions ($\sim 1\%$ elsewhere) moves its mean and late strides only from 0.049 and 0.030 to 0.056 and 0.033.

H2: the answer redirects the next question. *It barely does, and what response there is points the wrong way.* Pairing each answer with the stride into the following question, inside (puzzle, round-bin) cells so that the earlier arrival of a “yes” cannot manufacture the effect (Figure 2a; both the controlled and uncontrolled estimates, Appendix C): 4B $+0.001 [-0.009, +0.011]$, no detected response; 27B $+0.028 [+0.010, +0.046]$ and 397B $+0.009 [+0.002, +0.017]$, both responding with the sign inverted. Two distinct pathologies, and neither is the search a rational agent would run: the smallest tier does not hear the answer — nothing in the feedback channel can push it out of the hypothesis it started with, which is what circling is, now with a cause rather than a description.

And the direction is wrong, on a test that needs no premise about step size. Whether a refutation ought to produce a larger jump than a confirmation is not obvious: surprisal, belief change and distance in question space are different quantities, and “yes” is the rare answer here (18–26% of

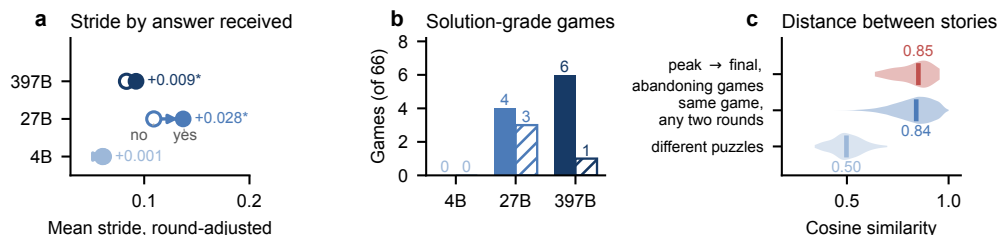


Figure 2: (a) Stride into the next question after a refutation (open) and a confirmation (filled), adjusted for round position. (b) Solution-grade games, and how many the agent ever volunteered. (c) In abandoning games the final story sits as close to the story that scored 1.00 as any two moments of the same game.

yes/no replies), so a searcher allocating steps by information might reasonably move further after one. The following comparison sidesteps the question: does the next question land nearer to regions the Oracle has *already refuted* or *already confirmed*? Told a “no”, all three tiers move toward what was already ruled out (-0.089 [$-0.112, -0.068$] for 4B, -0.051 [$-0.075, -0.032$] for 27B, -0.032 [$-0.047, -0.018$] for 397B, puzzle-clustered). Told a “yes”, no such asymmetry appears (all three CIs span zero) — which also answers the obvious objection, since refutations outnumber confirmations three to one in both cases. Moving back toward the excluded is wasteful under any account of search, and the smallest tier, whose stride ignores the answer, does it most strongly.

H3: when it finds the answer it will stop. *It does not stop.* Call a sustained peak ≥ 0.5 *solution-grade*. 4B reaches none; 27B reaches four and volunteers three; 397B reaches six and volunteers one (Figure 2b). Commitment is rare and late everywhere (11/66, 9/66, 2/66 games; mean round 11.1, 14.2, 23.5), and the agent often submits less than it had — 27B commits 0.075 on a game that peaked at 0.542. Under per-game criteria the tiers swap dominant failures: 4B circles in 39 of 66 games and abandons in 1, 397B circles in 6 and abandons in 12; none meets both.

Abandoning is a sub-semantic event, and that is why the geometry cannot score it. In abandoning games the final story sits at cosine 0.851 from the peak story that scored up to 1.00, while any two checkpoints of the same game sit at 0.841 and two stories from different puzzles at 0.498 (Figure 2c). The move that loses the answer costs 0.30 of score on average yet is indistinguishable, in this space, from ordinary within-game motion: a fatal substitution smaller than the encoder’s resolution cannot be predicted from that encoder.

Nor does the geometry predict the score. Drift does not appear (slopes cluster near zero everywhere; lexical novelty catches only 4B’s re-asks), and within tiers every stride and drift coefficient is bounded within ± 0.035 per SD (Appendix C): the geometry says *how* an agent fails, not how well it scores.

4 Limitations and Conclusion

Limitations. Everything here comes from one game, one Oracle protocol, and three models of one family; whether search under feedback behaves this way in other interactive settings is untested — the title’s “interactive LLM agents” is a hypothesis about scope, not a demonstration. The feedback measurement reads *how far* the next question moves, not *what* it asks, and round position is controlled only by binning. Truncated chain-of-thought under a tight token budget masquerades as an inability to form a question (§3), and the Oracle’s $\sim 95\%$ audited accuracy implies about one wrong answer per game. The anchor is not yet human-derived (§2), and the encoder’s resolution bounds what any embedding-based instrument can score (§3). One DeepSeek model serves as both Oracle and judge, and the tiers confound generation and architecture: model comparisons, not scaling laws. The difficulty measure may partly read familiarity (quantified in Appendix B; no claim rests on it). Puzzles involve death and dark themes; content-flagged.

Conclusion. These agents narrow without converging: the step shrinks four to five fold while the score stands still, and the feedback that should turn the search sends every tier back toward ground already ruled out — the smallest never finding the answer, the largest finding it and not stopping. The two failures need opposite fixes; one endpoint number scores them alike.

Declaration of LLM usage

LLMs are the object of study: the Questioner agents, the Oracle, and the causal-logic judge are all LLMs, specified in §3 and Appendix C, and every reported number derives from their logged interactions. Separately, LLM assistants were used to help write analysis code and edit prose; the authors reviewed all analyses, claims, and text, and take full responsibility for them.

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188 A Related work

189 Existing turtle-soup benchmarks, TurtleBench [1], SPLAT [2] (whose interactive player-judge pro-
 190 tocol we adopt), and TurtleSoup-Bench [3], score verification accuracy, solve efficiency, or faithful-
 191 ness to the intended story. TurtleSoup-Bench probes what agents ask next but grades it against the
 192 intended story; none of the three measures the search’s geometry, which is our delta. MUTATE [4]
 193 shows frontier agents find far fewer mechanism-distinct paths than humans; its diverge-then-narrow
 194 agent is a natural test intervention for our metrics. The instrument itself adapts forward flow [7]
 195 and the Divergent Association Task [8] from human psychometrics. The Small World of Words
 196 norms [9] provide the human anchor that ours is not yet (§2). Judge-bias catalogues [5, 6] shape
 197 the scoring: the model judge is confined to the one subscore that objective matching cannot reach
 198 (Appendix C).

199 B The puzzle set

200 **Where the puzzles come from.** A puzzle nobody has solved carries no evidence that its clues
 201 suffice, that its solution is unique, or that the intended leap is reachable; an agent scoring zero on
 202 it teaches us nothing. Our set is hand-picked from the classic repertoire of a Chinese Turtle Soup
 203 community: 22 puzzles people have played and solved, each keeping its source record. Every item
 204 was read and kept or rejected by hand; rejected were puzzles turning on deduction from stated
 205 evidence rather than a lateral reframing, and puzzles whose surfaces were incomplete at the source.
 206 Surfaces run 9–108 characters (median 34), solutions 13–200 (median 102).

207 **How hard are they?** Each puzzle carries an **under-determination** score: give a model the surface
 208 alone, with no Oracle and no feedback, let it write twelve complete stories, and take the closest to
 209 the real solution. The index is one minus that best guess, and is large when the surface leaves a
 210 solver far from the answer.

Under-determination	Puzzles
0.15–0.25	5
0.25–0.35	8
0.35–0.45	4
0.45–0.55	4
0.55–0.65	1

212 The median is 0.347, quartiles 0.273 and 0.410, full range 0.173–0.577: single-peaked with a thin
 213 hard tail. Easiest are the story the genre is named for (0.17) and the desert matchstick (0.19). Hardest
 214 is a fourteen-character surface reading *our heights differ / a flowerpot broke / our heights are the*
 215 *same* (0.58).

216 Two properties make this usable as a covariate. It describes the puzzle, not any agent’s performance
 217 on it. It is also independent of rounds used; a difficulty defined as “how many rounds this takes”
 218 would make an accuracy-versus-rounds plot confirm itself. Nor is it a proxy for surface length
 219 ($r = -0.22$, n.s.), so length dependence was not detected at $n = 22$ (a weak test). It is one measure
 220 rather than two: a surface loose enough to admit many mechanisms is, for that same reason, one
 221 whose true mechanism takes more reframing to reach, so *how far* and *how many ways* are not
 222 separable here.

223 **A caveat we report either way.** The two puzzles scoring easiest are among the most widely circu-
 224 lated, and the models’ own first-round guesses are consistent with familiarity contributing: on the
 225 best-known of them the two larger tiers open at 0.61 and 0.53 before any evidence arrives, against
 226 0.24 and 0.36 on a puzzle of almost identical under-determination. The measure may therefore read
 227 familiarity as much as difficulty on the easy end. Separating the two calls for a memorisation probe,
 228 which we have not run; nothing in this paper rests on the measure.

229 C Measuring agent performance

230 **What counts as a correct answer.** Judged outcome is **clue recall (70)** plus **causal-logic identity**
 231 **(30)**. Clue recall matches the answer against per-puzzle annotated key clues: objective and repro-
 232 ducible. The logic judge rates only whether cause $\rightarrow$ mechanism $\rightarrow$ outcome matches and is told

explicitly to ignore wording. Because single ratings are unstable on borderline answers, ratings are sampled and averaged (two samples per rating in the reported grid). The gap between the two subscores is itself readable: right story, wrong vocabulary.

Clue matching tolerates paraphrase through character-bigram recall, which sets a floor on how short an annotated clue may be. Below that floor only a verbatim match counts, and the score reads vocabulary rather than content. Clues are annotated above the floor.

Does the logic half earn its thirty points? On the full grid, yes: every model earns proportionally far more of the judged causal-logic half than of the clue-recall half (397B averages 12.6/30 logic against 7.2/70 clues, 27B 10.1 against 6.8, 4B 2.9 against 0.2), and the halves correlate only weakly per game (Spearman $\rho = 0.30/0.35/0.06$ for 397B / 27B / 4B over 396 E2 games each): models reach the right causal shape well before, and often without, landing the annotated clue content. The 70/30 weighting itself is a design choice; every conclusion in §3 is readable from the two halves separately, though the absolute plateau level (~ 0.2) does depend on the weights. The logic judge rates the causal-logic half on its 30-point scale and is sampled twice per rating at temperature 0.2; the two samples agree exactly in 86% of ratings and differ by under one point of 30 on average.

Feedback response: definition, confound, and both estimates. For each round t the Oracle’s answer is paired with s_{t+1} , the stride into the next question; “irrelevant” rounds are excluded. Confirmation arrives earlier than refutation (mean round 12.1 / 9.7 / 10.9 for a “yes” against 15.6 / 15.6 / 15.9 for a “no”) and early strides are larger, so an uncontrolled comparison is confounded by position. Pairs are formed inside (puzzle, round-bin) cells with bins {1–5, 6–10, 11–15, 16–20, 21–25, 26–29}, and 95% CIs bootstrap over puzzles (4,000 resamples). Uncontrolled, all three tiers appear to respond (+0.024 [+0.004, +0.044]; +0.043 [+0.015, +0.070]; +0.034 [+0.023, +0.046]); controlled, the smallest does not (+0.001 [−0.009, +0.011]; +0.028 [+0.010, +0.046]; +0.009 [+0.002, +0.017]). The gap between the two is the point.

The semantic scale of abandoning. Within abandoning games (sustained peak ≥ 0.3 , final below half of it; $n = 34$), the final checkpoint story sits at cosine 0.851 (median) from the peak story, while any two checkpoints of the same game sit at 0.841 (median over all 77,906 within-game pairs from 60 games) and two stories from different puzzles at 0.498 (231 pairs). The score gap across that same pair is $0.447 \rightarrow 0.099$.

The round-1 prior. The round-1 checkpoint precedes the first Oracle answer and is therefore a cold guess. It holds 49% / 33% / 26% (median, by tier) of the game’s eventual best score, and correlates with it at $\rho = +0.37/ +0.57/ +0.47$ (permutation $p = 0.09/0.003/0.03$, $n = 22$ puzzles with seeds averaged). Under-determination, itself measured from twelve cold guesses by a model cross-family with every Questioner, predicts round-1 quality at $\rho = -0.34/ -0.54/ -0.47$ ($p = 0.12/0.010/0.029$) but not the final score.

Behavioural indicators against puzzle structure. Regressing each per-trace indicator on under-determination, seeds averaged, 5,000 permutations: stride, late stride, drift (both anchors), mean anchor distance and rounds played are all n.s. at every tier ($|\rho| \leq 0.32$, $p > 0.10$). Commit rate is negative at all three (−0.32/ −0.25/ −0.43) but pooled $p = 0.13$, and 1 of 24 tests reaching $p < 0.05$ is the chance expectation, so we report it as unresolved rather than as an effect.

Retention estimator and permutation null. Retention is the mean per-game final-to-sustained-peak ratio over games with a nonzero sustained peak; the final is the last played checkpoint, and excluding early-committing games changes each tier’s value by ≤ 0.03 (bootstrap 95% CIs $\pm \approx 0.08$). For the null: unit, the game. For each of 2,000 permutations, every game’s checkpoint sequence is shuffled within the game (knowledge-constant, jitter-only null), the sustained peak and the final (last element) are recomputed on the shuffled sequence, and retention is the mean over that tier’s games with a nonzero permuted peak. The reported p is one-sided per tier, the fraction of permutations whose null retention is at or below the observed value. Observed retention (0.69–0.76 by tier) does not fall significantly below this null (0.79–0.84, $p \geq 0.09$), so the peak-to-final gap visible in Figure 1b is reported as consistent with scoring jitter and no claim in the body rests on its size; the commitment record of §3 is behavioural and unaffected by jitter. Late-game stride means the second half of a game’s strides.

Mixed-model coefficients. Best checkpoint accuracy per game; Gaussian LMM, puzzle random intercept, tier fixed effect, ML fits, likelihood-ratio tests against the covariate-only model. Stride

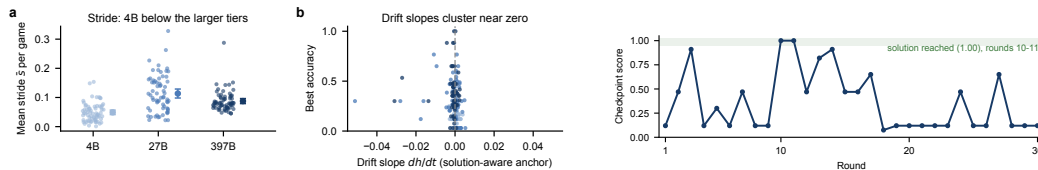


Figure 3: Per-trace geometry ($n = 198$). (a) Per-game mean stride by tier (dots; bars, tier mean $\pm$ 95% CI): 4B sits below the larger tiers. (b) Drift slope against best accuracy, clustered near zero at every tier.

Figure 4: One abandoning game (refsoup_021, 397B, seed 2): the round 10–11 checkpoint stories are the solution (band), never volunteered, abandoned by round 19. Full rows in Appendix C.

287 $\beta = -0.010$ per SD, 95% CI $[-0.035, 0.014]$, $p = 0.42$; drift $\beta = -0.011$, $[-0.035, 0.014]$,
 288 $p = 0.39$; stride $\times$ tier $p = 0.20$; surface-anchor drift $\beta = -0.020$, $[-0.044, 0.003]$, $p = 0.097$.

289 **Environment settings and Oracle audit.** Every model call uses max_tokens 2048 (thinking dis-
 290 abled, temperature 0.2) with a per-game budget of 50,000 tokens; exhausting the game budget forces
 291 a final answer and is recorded as the game outcome token_budget. The Oracle audit probes a
 292 candidate with 25 hand-crafted (question, expected-answer) pairs over three puzzles, 21 with a defi-
 293 nite yes/no answer and 4 expecting “irrelevant”; the pass bar is 90% on the yes/no items. The
 294 deployed DeepSeek-V3.1 scored 20/21 (95%) on yes/no and 3/4 exact on “irrelevant”. Candidates
 295 measured in the pilot phase: a 4B Oracle with thinking disabled, 30% (rejected; in that pilot config-
 296 uration, raising Oracle accuracy from 30% to 100% left the 4B Questioner’s score unchanged); the
 297 same model with thinking enabled, 70% (rejected); a same-family flash model, 90% (excluded for
 298 sharing the Questioners’ family); GLM-5.3-flash, 100% (used in earlier tooling).

299 **Trajectory and difficulty pipeline.** Keywords are extracted with jieba after stripping Latin char-
 300 acters and digits; each keyword is embedded with bge-small-zh-v1.5 (normalized), and the round
 301 vector is the re-normalized mean of its keyword embeddings. All distances are cosine. The solution-
 302 aware anchor embeds the puzzle’s surface, solution, and clue terms; the surface-only anchor embeds
 303 surface terms alone. The under-determination measure prompts GLM-5.3-flash (cross-family with
 304 all Questioners) for twelve cold guesses from the surface and scores closeness as embedding simi-
 305 larity between guess and solution under the same encoder, not with the composite score, so the
 306 covariate is not coupled to the outcome apparatus.

307 **Artifacts.** Harness, puzzle set with per-item source records, per-game logs (including every check-
 308 point story), and the analysis scripts behind each reported number are released at a public repository,
 309 withheld here for double-blind review and available to reviewers on request through the programme
 310 chairs. Puzzle surfaces and solutions originate from a public community repertoire and are redis-
 311 tributed with source attribution and a content flag.

312 **Where the accuracy curve comes from.** E1 does not score once at the end. Every round, after
 313 the Oracle answers, the Questioner is required to commit its best complete story, and *that* is scored.
 314 Each point on the accuracy curve is therefore a full answer written with the evidence available at
 315 that round.

316 **One game, worked.** Puzzle refsoup_021 (surface: a stormy night, a run-down temple, two
 317 photographs on the wall that seem to stare; by morning the narrator sees only two windows; solution:
 318 “there were never photos on the wall; people were pressing against the windows, looking in”), played
 319 by Qwen3.5-397B, seed 2. Table 1 shows seven of thirty rounds; Questions and stories are our
 320 translations of the logged Chinese, stories abridged to the clause carrying the hypothesis. Rounds
 321 22–30, eight lexically distinct re-asks of the same flooding hypothesis (e.g. round 25, whether the
 322 narrator woke floating upside down; round 28, whether rising water reached eye level), are in the
 323 released logs.

324 Three things the table shows that no mean curve can. **The checkpoint mechanism surfaces knowl-**
 325 **edge the agent never acts on.** The round-10 committed story *is* the solution, synthesized from
 326 earlier answers on a round whose own question was answered “no”. Yet the model never volun-
 327 teers a final answer in all thirty rounds (one of the 57/66 non-committing 397B games), and by
 328 round 19 it has abandoned the correct reading for a falling-coverings mechanism it never escapes.
 329 **Stalling in fresh words happens at the largest scale too.** Rounds 22–30 are eight lexically distinct

Rd	Question (abridged)	Ans.	Committed story (abridged)	score (clue+logic)
1	Are the ones in the photos people?	yes	The two “photos” are two people nailed or sealed into wall niches.	0.12 (0+12)
3	On waking the next day, is he still in the temple?	yes	The “two photos” are really two windows, mistaken for photos in the dim light.	0.91 (70+21)
4	Are the “two windows” the same objects as last night’s “photos”?	yes	The “photos” were hollowed-out window openings; the fear came from darkness and illusion.	0.12 (0+12)
10	Did frame glass break and fall, leaving open holes?	no	They were never photos, but faces pressed against the outside of the windows.	1.00 (70+30)
11	Were the photos removed, revealing windows behind?	no	The staring figures were people outside the windows.	1.00 (70+30)
19	Were real people hiding behind the photos?	no	Window openings sealed by painted cloth covers; the storm tore them off.	0.12 (0+12)
30	Did rising water lift the narrator higher?	no	Papered-over window openings; the storm stripped the paper away.	0.12 (0+12)

Table 1: Seven rounds of one E1 game (refsoup_021, Qwen3.5-397B, seed 2): the Questioner’s question, the Oracle’s answer, the story the agent was forced to commit that round, and its composite score. Questions and stories are our translations; the original transcripts are in the released logs.

330 re-asks of the same flooding hypothesis, exactly the pattern lexical novelty misses. **The curve’s**
331 **round-to-round jitter is a scoring fact, not a knowledge fact.** The same understanding re-worded
332 (rounds 3 → 4) swings the clue half 70 → 0. The mean curves of §3 average this out, which is why
333 single-game curves are not the unit of analysis.